

Lecture 22: Time Series Models

*Instructor: Jonathan Pipping-Gamón**Author: MS*

22.1 When Order Matters

Every model in this course so far has treated observations as exchangeable. A regression for expected goals does not care which shot came first; a kernel density estimate would look identical if we shuffled the rows of the data. That assumption is convenient, and for a single season of shot data it is reasonable: one shot does not directly cause the next.

Many sports quantities do not behave this way. Suppose someone asks: *how many three-pointers will the NBA have attempted after one season? After two? After three?* This is a forecasting question about a running total, and running totals only ever go up — they are a cumulative sum of a seasonal process. Figure 22.1 plots the league's cumulative three-point attempts, season by season, from 2010–11 through 2023–24.

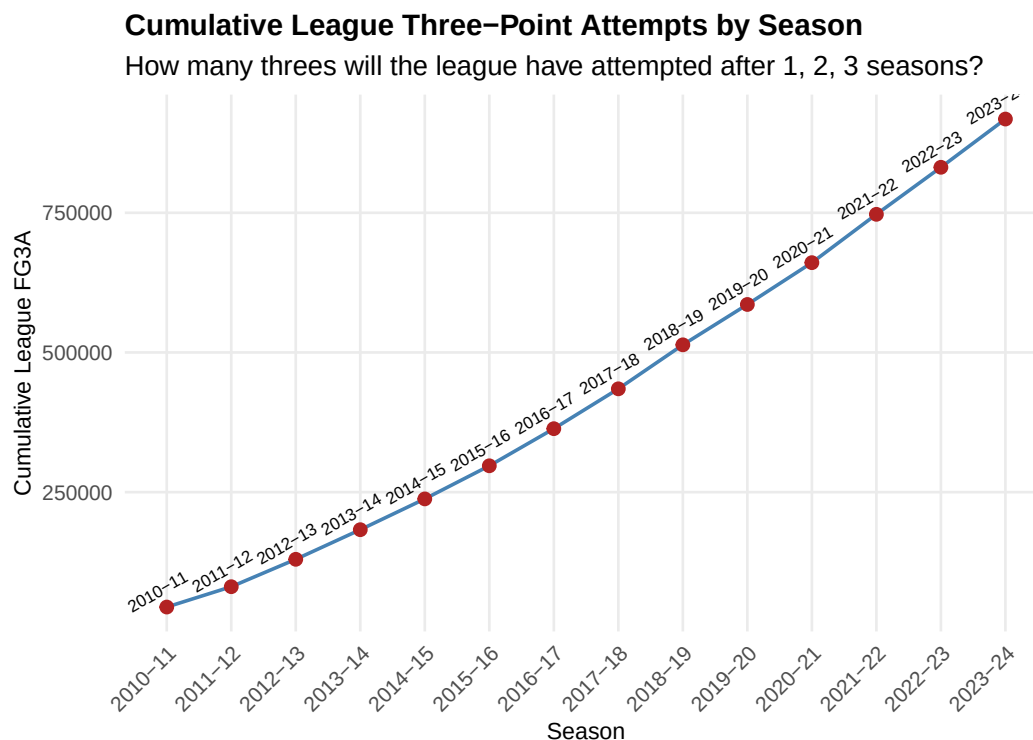


Figure 22.1: Cumulative league-wide three-point attempts (FG3A) after each season, real NBA data 2010–11 through 2023–24.

A cumulative sum is the simplest possible example of a non-stationary series: its mean is, by construction, increasing in t . To forecast it well, the natural object to model is the *seasonal increment* — how many threes

were attempted in a given season — since the cumulative total is recovered by summing those increments back up. Figure 22.2 plots that series directly: league-wide total three-point attempts per season.

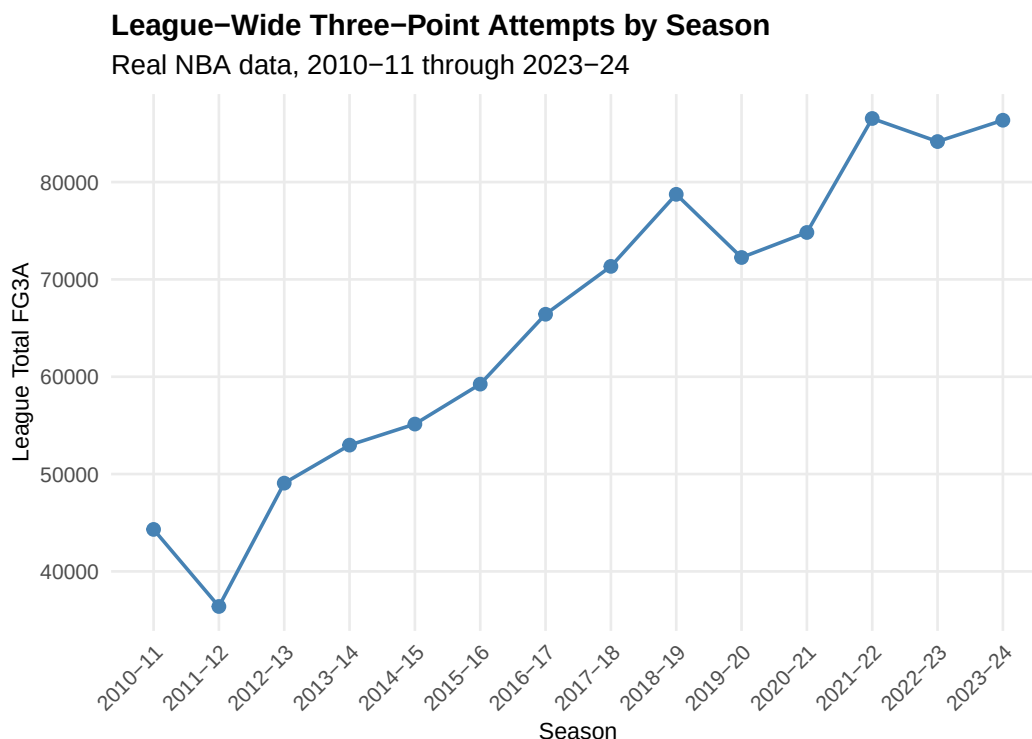


Figure 22.2: League-wide total three-point attempts by season, real NBA data. The upward trend reflects the well-documented three-point revolution.

The trend is obvious and sustained. Our question for today: if Y_t is the league's total FG3A in season t , what can we say about Y_{t+1} using the history of Y itself?

22.2 Autocorrelation and the ACF

Before we build any model, we need a way to measure whether the past predicts the present.

Definition 22.1 (Autocorrelation). *For a time series $Y_1, \dots, Y_T$, the lag- k autocorrelation is the correlation between the series and a copy of itself shifted k periods,*

$$\rho_k = \text{Corr}(Y_t, Y_{t-k}).$$

When ρ_k is far from zero, knowing Y_{t-k} tells us something about Y_t . When $\rho_k \approx 0$, the value k seasons ago carries no linear information about this season.

The **autocorrelation function (ACF)** plots $\hat{\rho}_k$ against k for $k = 1, 2, 3, \dots$. Dashed lines mark the approximate 95% confidence band: bars that extend beyond them are statistically distinguishable from zero. Figure 22.3 shows the ACF for the raw league three-point series. Because we have only 14 seasons, we cap the lags at 7.

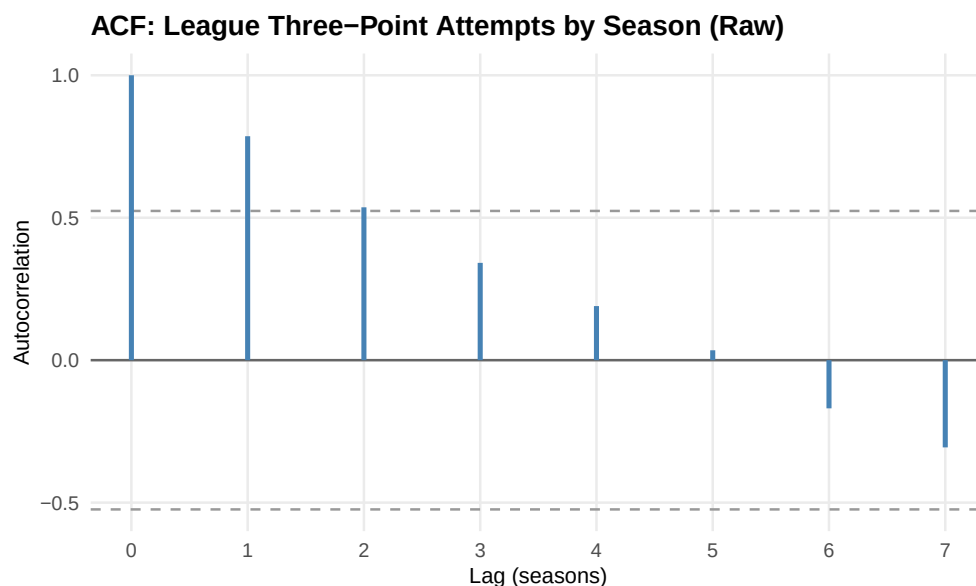


Figure 22.3: ACF of the raw league three-point attempt series (seasonal). Bars outside the dashed band are statistically distinguishable from zero at the 95% level.

The bars decay slowly and stay positive across all available lags. This is the classic signature of a trending, non-stationary series. We will return to what that means — and how to fix it — in Section 22.6.

22.3 The PACF: Removing What the Middle Already Explains

The ACF has a limitation. Suppose Y_t depends only on Y_{t-1} , but Y_{t-1} depends on Y_{t-2} . Then Y_t will be correlated with Y_{t-2} purely through the chain running via Y_{t-1} — even though Y_{t-2} adds nothing once we already know Y_{t-1} . The ACF cannot distinguish direct dependence from indirect dependence; it shows the total correlation at each lag regardless of what is driving it.

Definition 22.2 (Partial autocorrelation). *The lag- k partial autocorrelation ϕ_{kk} is the correlation between Y_t and Y_{t-k} after removing the linear effect of everything in between, $Y_{t-1}, \dots, Y_{t-k+1}$.*

The **partial autocorrelation function (PACF)** plots ϕ_{kk} against k . It answers a sharper question: does lag k matter on its own, or only because it correlates with more recent lags?

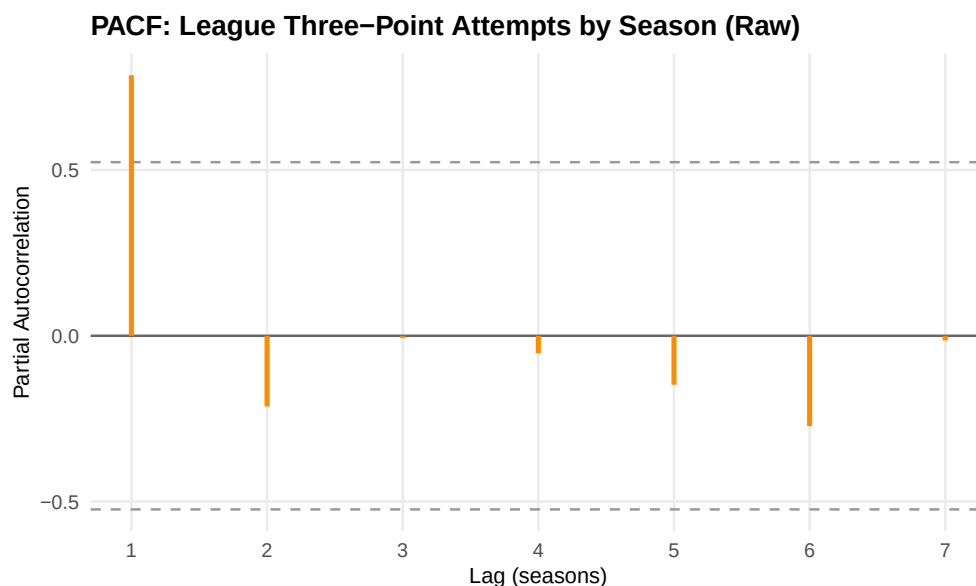


Figure 22.4: PACF of the raw league three-point attempt series (seasonal).

Together, the ACF and PACF form the main diagnostic for selecting a model. A useful rule of thumb, which we return to once our series is stationary:

- **PACF cuts off** sharply after lag p , ACF decays gradually $\Rightarrow$ $\text{AR}(p)$ is a natural candidate.
- **ACF cuts off** sharply after lag q , PACF decays gradually $\Rightarrow$ $\text{MA}(q)$ is a natural candidate.
- Both decay gradually $\Rightarrow$ ARMA is usually needed.

22.4 Autoregressive Models

An autoregressive model says this season's total is a linear function of recent seasons plus noise,

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t.$$

The coefficient ϕ_i measures how much weight the value i seasons ago carries today. When $|\phi_1|$ is close to one, observations change slowly and consecutive seasons are highly correlated. When ϕ_1 is near zero, last season provides almost no information beyond the long-run mean.

Figure 22.5 illustrates these two extremes with simulated series.

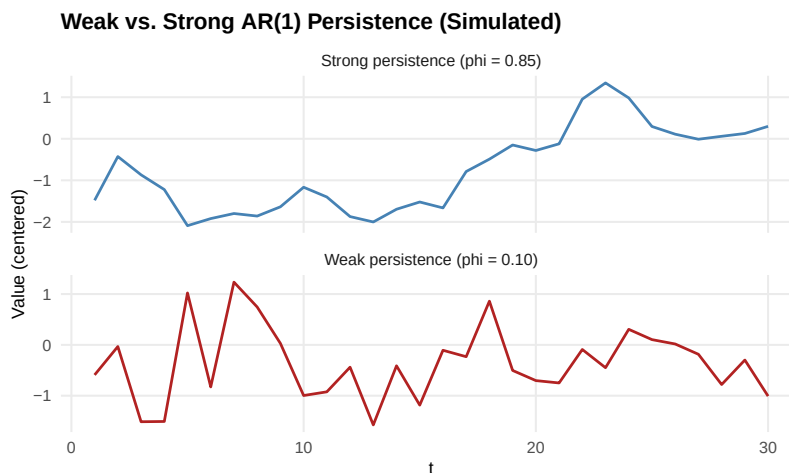


Figure 22.5: Simulated AR(1) series: weak persistence ($\phi = 0.10$, top) looks close to noise; strong persistence ($\phi = 0.85$, bottom) produces smooth, slowly-meandering swings.

The PACF is the diagnostic that tells us p : a sharp cutoff after lag p with little structure beyond it suggests AR(p). Fitting AR(1) to the differenced league series (introduced in Section 22.6) gives $\hat{\phi}$, quantifying how much of one season's deviation from trend persists into the next.

22.5 Moving Average Models

The phrase “moving average” is misleading — this is *not* a rolling average of past observations. It is a model of how recent *forecast errors* echo forward.

Suppose the league was expected to attempt a certain volume this season, but a rule change or sudden style shift drove attempts far above expectation. That shock — a large positive ε_{t-1} — may continue to influence the following season even after the immediate cause has been absorbed. An MA model captures this lingering effect:

$$Y_t = \mu + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}.$$

Unlike AR models, MA models have no memory beyond q periods: a shock's influence vanishes completely after q steps rather than decaying gradually. This is why the ACF — not the PACF — cuts off sharply at lag q for a pure MA(q) process, and why the ACF–PACF pairing in Section 22.3 is the key diagnostic tool.

22.6 The I: Stationarity and Differencing

AR and MA models both assume the series is **stationary** — that its mean, variance, and autocorrelation structure do not change over time.

Definition 22.3 (Weak stationarity). *A series Y_t is weakly stationary if $E(Y_t)$ is constant in t , $\text{Var}(Y_t)$ is constant in t , and $\text{Cov}(Y_t, Y_{t-k})$ depends only on the lag k .*

Our raw series is not stationary: its mean has grown every season for over a decade. The augmented Dickey–Fuller (ADF) test checks for a unit root and fails to reject the unit-root null on our raw series, consistent with what the slowly decaying ACF already suggested. (With only 14 seasons the test has limited power; treat it as corroborating evidence alongside the visual inspection.)

The fix is **differencing**: instead of modeling Y_t directly, we model the season-over-season change,

$$\nabla Y_t = Y_t - Y_{t-1}.$$

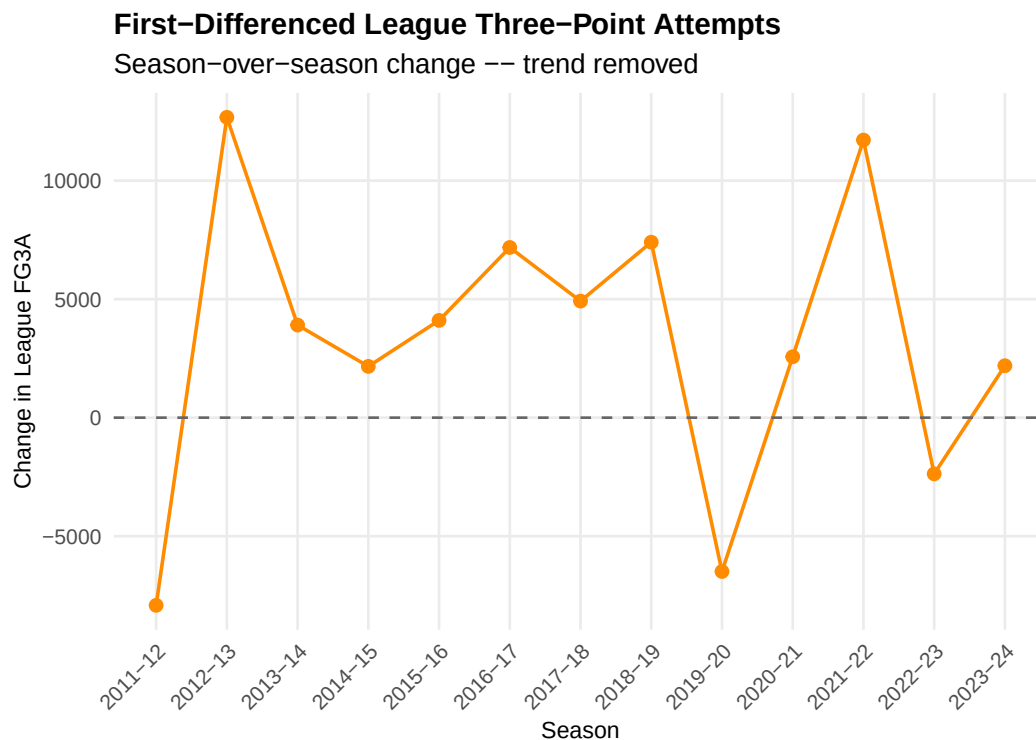


Figure 22.6: First-differenced league three-point attempts (season-over-season change). The long-run trend is removed; the series now fluctuates around zero.

After differencing, the ADF test rejects the unit-root null, and the ACF and PACF of the differenced series (Figure 22.7) decay quickly — evidence that AR and MA models are appropriate on ∇Y_t .

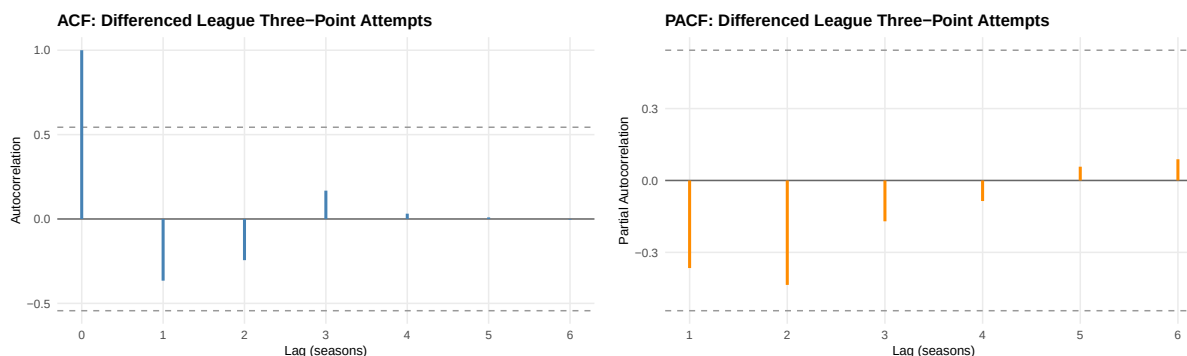


Figure 22.7: ACF (left) and PACF (right) of the first-differenced series. Both decay quickly, consistent with stationarity.

Note the connection back to Section 22.1: differencing the cumulative series recovers the seasonal series, and differencing the seasonal series produces these season-over-season changes. `diff()` and `cumsum()` are inverses up to the initial value.

22.7 ARMA and ARIMA

An $\text{ARMA}(p, q)$ model combines AR and MA on a stationary series,

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}.$$

When the raw series is not stationary, we difference first and then fit ARMA to ∇Y_t . This three-piece combination is $\text{ARIMA}(p, d, q)$: AutoRegressive, Integrated (the differencing step), Moving Average. Here $d = 1$ since one difference was sufficient to achieve stationarity.

We select p and q by comparing information criteria across candidate models (Table 22.1).

Model	AIC	BIC
AR(1)	294.2	296.1
MA(1)	303.6	305.5
MA(2)	301.9	304.4
ARIMA(0,1,0) — random walk with drift	266.6	267.7

Table 22.1: Model comparison on the league three-point attempt series (seasonal, 2010–11 through 2023–24). Lower AIC/BIC indicates a better trade-off between fit and complexity. $\text{ARMA}(p, q)$ with $d = 0$ selected AR(1) as the best no-differencing model; $\text{ARMA}(1, 0)$ is identical to AR(1) and is not listed separately.

The winner by a large margin is $\text{ARIMA}(0, 1, 0)$ — a **random walk with drift**. This model says the best forecast for next season’s three-point attempts is simply this season’s total plus a constant annual increment. No AR or MA terms are needed once we difference out the trend.

This is a meaningful result, not a failure. The three-point revolution from 2010 to 2024 has been nearly monotone: season-over-season growth has been so consistent that past *deviations* from trend carry almost no

information about future deviations. The trend itself is the signal. Once it is removed by differencing, the residual series is close to white noise — and $\text{ARIMA}(0,1,0)$ captures exactly that.

Figure 22.8 shows the forecast for the next three seasons.

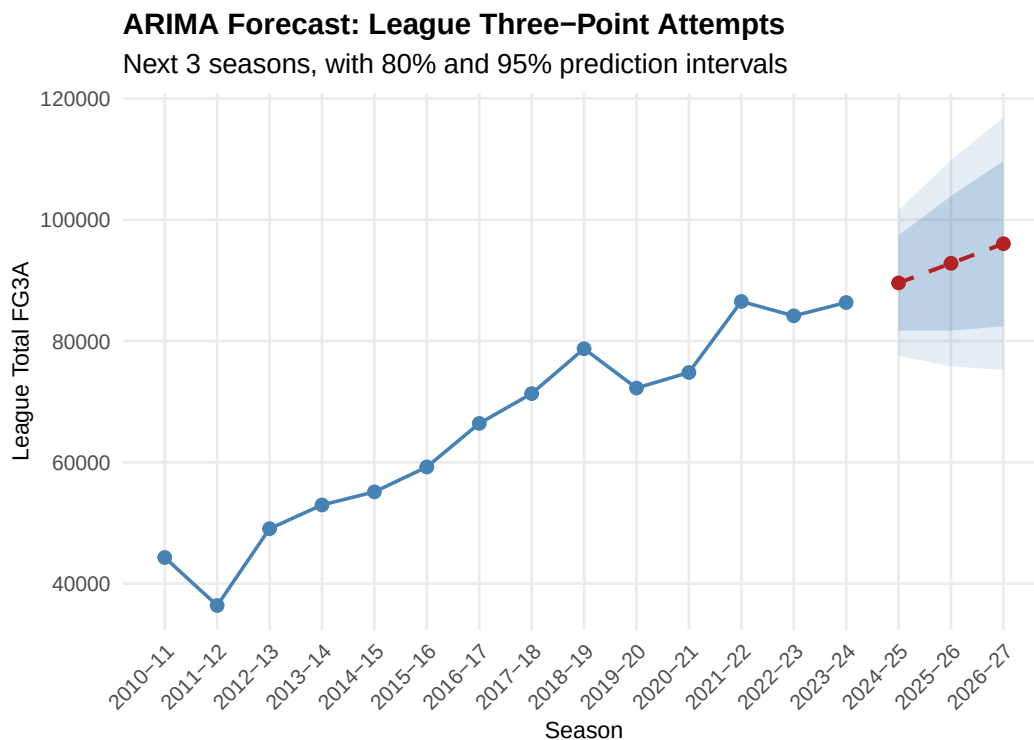


Figure 22.8: ARIMA forecast for league-wide three-point attempts, next three seasons, with 80% and 95% prediction intervals.

22.8 Takeaways

- Time series data violate the IID assumption; the ACF and PACF are the primary diagnostic tools.
- The **ACF** measures total correlation at each lag including indirect paths; the **PACF** isolates the direct contribution of each lag after partialling out everything in between.
- A slowly decaying ACF that stays positive across all lags signals non-stationarity — typically a trend.
- **AR** models build memory through past *values*; the PACF cutoff identifies the order.
- **MA** models build short-lived memory through past *forecast errors*; the ACF cutoff identifies the order.
- **Differencing** removes a trend and is not itself a model — it is a preprocessing step that makes AR and MA valid. `diff()` and `cumsum()` are inverses.
- **ARIMA**(p, d, q) combines all three pieces; p and q are selected by AIC or BIC.
- With small samples ($n = 14$ here), formal tests have limited power. Visual inspection of the ACF and PACF should carry at least as much weight as any p -value.

Looking ahead: today's models use only the history of Y_t itself. Real forecasting also wants external information — number of games played, rule changes, team-level covariates — which motivates dynamic regression (ARIMAX) models, and eventually state-space models where a latent “league shooting environment” evolves as a hidden variable over time.